

Experimental and numerical investigation of the aerosol generation mechanisms at the human vocal folds

Lisa Fritzsche^{*1}, Martin Heinrich¹, Kartik Gupta¹, Rüdiger Schwarze¹, Katrin Bauer¹

¹Institute of Mechanics and Fluid Dynamics, TU Bergakademie Freiberg,
Lampadiusstraße 4, 09599 Freiberg, Germany

*Corresponding author: lisa.fritzsche@imfd.tu-freiberg.de

Abstract

Aerosolized droplets emitted during respiratory activities play an essential role in spreading various viral respiratory infections. Although it is widely accepted that large amounts of these droplets are created by upper airway activity, the sites and mechanisms of aerosol generation remain unclear. Our pioneering study focuses on the atomization process of airway mucus at the vocal folds (VF). We used a simplified in vitro model of the human VF that emulates their oscillating movement during phonation. The primary atomization of an artificial mucus applied to the VF was observed using optical measurement techniques. This novel insight shows how the process of aerosol generation is dominated by the burst of bubbles spanning the glottis. Furthermore, we analyzed the influence of the different phonation parameters on the resulting droplet size distributions. Additionally, three-dimensional multiphase CFD (computational fluid dynamics) simulations were conducted to get a deeper insight into the formation and breakup of liquid bridges at the VF. These simulations allowed us to use a more realistic geometry and implement wetting and boundary conditions in a more physiologic manner than in the experimental setup. The extension and breakup of liquid films proceeds quite similar to what can be observed in the experiments.

Keywords

Vocal folds, bubble breakup, aerosol, respiratory droplets, airway mucus atomization, optical measurements, CFD

Introduction

Throughout the last century, it was generally assumed that contagious respiratory diseases primarily spread among people through short-range droplet transmission, whereby large respiratory droplets ($> 50 \mu\text{m}$), released by an infected person during coughing and sneezing, come into contact with the recipient's mucous membranes [1]. However, such large droplets fall to the ground within a few seconds and within a short distance of one to three meters [2]. Only during the past two decades, and particularly due to the recent COVID-19 (Coronavirus Disease 2019) pandemic, has awareness emerged regarding the immense danger posed by respiratory aerosols, which are emitted in large quantities even during normal speech and breathing. These aerosol particles are so small that they can accumulate in the ambient air, invisible to the naked eye, and remain suspended for hours to days, yet they are still large enough to carry various pathogens [3, 4]. When inhaled, these particles are able to penetrate deep into the respiratory system due to their small size, infecting the new host [3]. As potentially infectious droplets released during exhalation can be formed throughout the respiratory system, from the alveoli to the mouth/nose, there is a particular need for research on the mechanisms and sites of aerosol generation [5].

To date, no studies are known regarding the formation of aerosols within the upper respiratory tract. However, high viral loads in throat swabs of COVID-19 patients during the early symptomatic phase indicate viral replication in the tissue of the upper respiratory tract [6]. Also, Asadi et al. [7] hypothesized that vocal fold vibration is an essential mechanism of aerosol generation. As particles of critical sizes (droplet nuclei of $\mathcal{O}(1 \mu\text{m})$) for the airborne transmission of pathogens are generated during vocalization, there is a current and pressing interest in investigating the atomization of respiratory liquid (mucus) at the human vocal folds [5, 8]. The aim of this work is, therefore, to provide a first piece of the puzzle on the way to understanding the mechanisms behind the production of aerosol particles during phonation and to identify the key factors that significantly impact their size distributions. Yet, absolute particle numbers are not in the focus of interest within the scope of this study. To investigate the mechanisms of mucus breakup during sustained phonation, we conducted experiments at a simplified in vitro VF model as well as three-dimensional (3D) multiphase CFD simulations using a more realistic VF geometry. Additionally, using the experimental model, we are able to determine particle size distributions and examine the impact of oscillation frequency, amplitude and exhalation air flow rate.

Experimental material and methods

The experimental setup and techniques are described in detail in a previous publication by the authors (Fritzsche et al. [9], 2022). The schematic measurement setups are summarized in Figure 1. As shown in Figure 1(a), humidified air enters the simplified trachea, a perspex duct with rectangular cross section, and exits through the VF model, which is attached to the upper end. The model represents the human vocal folds (see Fig. 1(a),(b)) by dividing their complex shape into a rigid, physiologically contracting part ("flow guides"), and two elastic silicone cylinders ("vocal folds"). The main dimensions of the overall VF model, particularly the shape of the flow guides, are based on the M5 model introduced by Scherer et al. [10]. The silicone cylinders have a length of 20 mm and a diameter of 4 mm to match physiological values [11]. To replicate the elastic properties of VF tissue, a highly

flexible and soft silicone material (KauPo EcoFlex® 00-20), with a Young's modulus of 0.06 N/mm^2 , is used [12, 13]. In order to modify the silicone's surface properties, gas phase fluorination was employed, improving the wettability and mitigating the materials' stickiness.

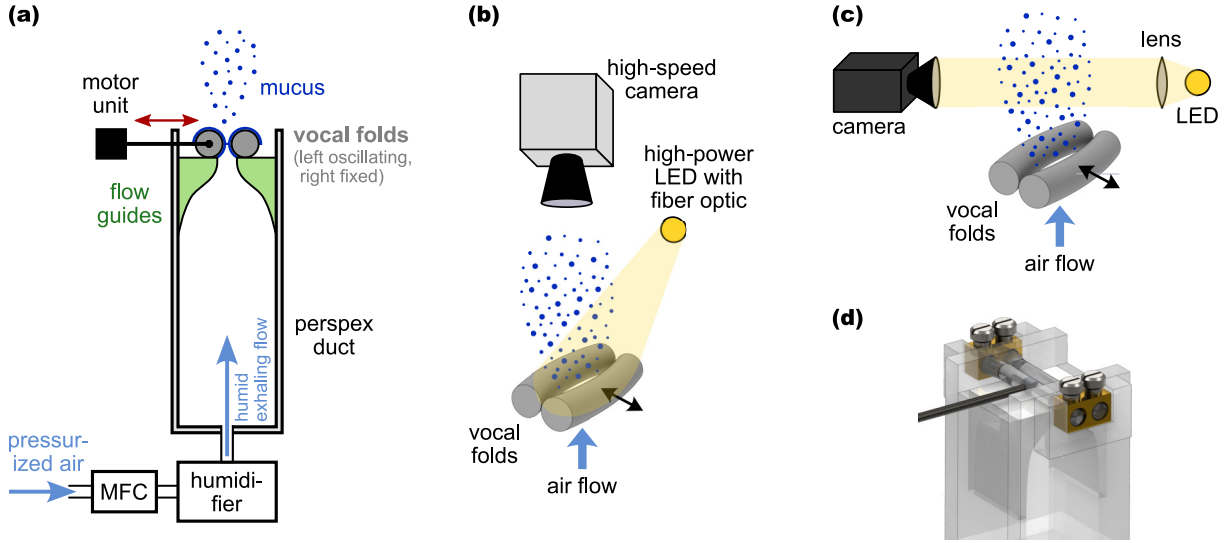


Figure 1. Experimental setup: (a) schematic representation of the test rig with the main components, (b) setup for recording the VF motion and the liquid film breakup in top view, (c) shadowgraphy setup for particle size measurements, (d) CAD rendering of the uppermost part the flow channel with the VF model.

To gain full control over the oscillation parameters, i.e., being able to tune frequency and amplitude freely and separately, the vocal folds are agitated by a motor rather than aerodynamically by the respiratory airflow as it would be the case with in vivo VF. The motor's rotational motion is converted into a translatory motion through an eccentric unit connected to the VF by a rigid threaded rod, which results in a sinusoidal VF motion. For now, only one vocal fold is actively oscillated while the other remains fixed. Despite this limitation, the opening and closing mechanism of the model VF is considered sufficient to emulate the physiological glottal cycle for the purposes of this investigation. Based on the findings of Padday et al. [14], we expect no significant impact on the breakup of liquid bridges.

The oscillation frequency f of the VF is varied between 75–150 Hz with amplitudes $\hat{x}$ of 0.5, 0.7 and 1.0 mm (resulting in glottal opening gaps of 1.0, 1.4 and 2.0 mm). Thus, the VF oscillation parameters are adjusted according to the typical human vocalization range, which exhibits maximum glottal gaps of $\mathcal{O}(1 \text{ mm})$ and fundamental frequencies between 80–200 Hz [15]. To replicate typical exhaling flow rates [16], a mass flow controller (MFC) ensured a constant air flow rate $\dot{V}$ at 10, 20 and 30 l/min. The exhaled air had a temperature of 23 °C and relative humidity (RH) of about 75 %.

During the experiments, a syringe pump was used to continuously feed the surface of the VF with a model respiratory liquid. To assure constant wetting of the VF without overflow, the mucus flow rate was arbitrarily set to 10 ml/h. The model liquid (DMEM GlutaMAX with 5 % bovine serum) exhibits shear-thinning and thixotropic behavior to emulate the non-Newtonian characteristics of respiratory mucus. The rheological properties of this artificial mucus were measured via a rotational rheometer (cone-and-plate system) and can be fitted using a Power Law, which is also used to model the mucus phase in the numerical simulations and thus described in detail in equation (8). It has to be noted that the artificial mucus is less viscous than tracheal mucus [17]. Yet, the properties of laryngeal mucus from the VF remain largely unknown, as examining it is exceedingly challenging due to the very limited sample size and hence investigations were focused solely on viscoelasticity so far [18]. The model liquid's surface tension value of about 50 mN/m lies within a physiologically reasonable range [19, 20].

Two separate setups were used to conduct two different measurement procedures (details in the following sections): one to visualize the atomization processes and the other to measure particle size distributions. A combined analysis of the results will unveil the processes responsible for the emergence of different droplet sizes and the impact of the exhalation parameters on mucus breakup and droplet size distribution.

Atomization process visualization in a top view of the vocal folds

To study phenomenological aspects, the VF movement and the mechanisms of mucus breakup at the vocal folds are recorded using a high-speed camera (Phantom VEO 710L) over the course of several (≈ 50 per case) oscillation periods. A schematic illustration of the setup is shown in Figure 1(b). With a framerate of 5000 fps, it is observed how liquid films and bridges are formed and subsequently break apart into droplets. High intensity and short exposure time (60 μs) illumination is provided by a focused high-power LED (light emitting diode) in order to minimize motion blur.

Particle size measurements using a shadowgraphy setup

To quantitatively analyze particle sizes resulting from the atomization process, we directly observed the mucus spray above the VF at a distance of 25 mm using a shadowgraphy setup (see Fig. 1(c)). For illumination, the light of an LED was focused and parallelized by a short focal-length Fresnel lens. By backlighting the droplets with an LED flash, their shadows were imaged on the sensor of a FLIR Grasshopper3 camera (4 MP resolution). To ensure particle images without motion blur, the LED flash had an extremely short duration of 2 μ s. The frame rate of 34 fps was set lower than the VF frequency to prevent duplicate recordings of the same particle in subsequent images. With a resolution of 2.6 μ m per pixel, the setup provided a total field of view of 5.3 mm $\times$ 5.3 mm and a depth of field of approximately 1.5 mm. This measurement volume comprised a representative section of the spray. To ensure statistically reliable results, approximately 8000 images were acquired for each parameter set.

After the acquisition, the images are evaluated in the following way: Initially, frames without particles are removed to decrease computation time, leaving approximately 5000 images per measurement case. In-house developed PYTHON software is employed to detect and measure the sizes of imaged droplets in each of the remaining images. Figure 2 depicts the fundamental steps of the image evaluation for particle detection. Since some mucus droplets cause local darkening by hitting the Fresnel lens, the evaluation is performed on intervals of 50 consecutive images. The background is considered constant within each interval and a mean image is created to remove the background from the raw images through division. The resulting gray values are rescaled to 8-bit image color depth (0–255). To prevent distorted particle sizes caused by particles being out of focus, trinarization, i.e., applying two different thresholds to the image, is performed. These threshold values are determined through test evaluations to ensure accurate detection of in-focus particles based on visual assessment. The software generates binary images using the two threshold levels, detects all particles in them and stores, among others, the particle sizes and positions. Assuming perfectly spherical particles, an equivalent diameter is derived from the number of pixels per particle. The diameter difference between the upper and lower threshold is calculated for each particle as a criterion to determine the maximum allowable blur. If the diameter difference exceeds a certain specified value, indicating significant variation in size between thresholds, the particle is considered blurred and is therefore discarded, as out-of-focus particles have distorted sizes in the image, making it impossible to accurately determine their true dimensions. In contrast, particles in focus show minimal to no size variation when the thresholds are applied. The program then calculates the real particle diameters in micrometers based on the image resolution. Additionally, the program calculates an eccentricity value for each particle as a measure of circularity, confirming that most detected particles are nearly perfectly spherical.

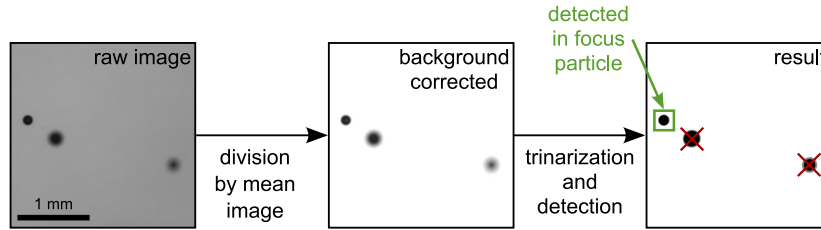


Figure 2. Steps of determining particle sizes, visualized by means of a section of an exemplary shadowgraphy image. The resulting image displays the superposition of two binary images that were created using different threshold values (trinarization). For differentiation, the particles in one binary image are shaded in gray and black in the other. When particles are in focus, their size does not change by applying different thresholds.

In the subsequent postprocessing step, particle size distributions are visualized and analyzed further. The particle size data undergo log transformation (base 10) and then a kernel density estimator (KDE) with Gaussian kernel is used to predict the underlying probability density function (PDF).

Numerical setup

The liquid bridge breakup between the vocal folds was simulated with the open-source CFD library OpenFOAM v2212 [21]. The simulations were performed at the Center for Information Services and High Performance Computing (ZIH) at TU Dresden, Germany.

Governing equations

The physics of multi-phase flow inside the vocal folds is governed by the three-dimensional, incompressible Navier-Stokes equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (1)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \mu \Delta \mathbf{u} + \rho \mathbf{g} + \mathbf{f}_\sigma \quad (2)$$

with the fluid density ρ , fluid velocity $\mathbf{u}$, static pressure p , dynamic viscosity μ , gravitational acceleration $\mathbf{g}$, and surface tension force $\mathbf{f}_\sigma$. In order to account for the two-phase laminar flow of air and mucus, the model is extended for two incompressible, immiscible fluids using the Volume-of-Fluid method with the *isoAdvector* geometrical interface

capturing and advection method developed by Roenby [22]. Here, an additional equation for the phase fraction α is solved with:

$$\alpha = \begin{cases} 0 & \text{within the air phase} \\ 0 < \alpha < 1 & \text{interface} \\ 1 & \text{within the mucus phase} \end{cases} . \quad (3)$$

The surface tension force $\mathbf{f}_\sigma$ in the momentum equations is formulated as volumetric force according to the continuum surface force (CSF) model of Brackbill et al. [23]:

$$\mathbf{f}_\sigma = \sigma k \mathbf{n} \quad (4)$$

with the interface normal vector $\mathbf{n}$ and free surface curvature k calculated as:

$$\mathbf{n} = \frac{\nabla \alpha}{|\nabla \alpha|}, \quad k = -\nabla \cdot \mathbf{n} . \quad (5)$$

Material properties for both phases in the simulation are oriented toward the experimental values and calculated linearly based on the phase fraction:

$$\rho = \rho_{\text{air}}(1 - \alpha) + \rho_{\text{mucus}}\alpha , \quad (6)$$

$$\mu = \mu_{\text{air}}(1 - \alpha) + \mu_{\text{mucus}}\alpha . \quad (7)$$

Air is modeled as Newtonian fluid with constant density and viscosity of $\rho = 1.186 \text{ kg/m}^3$ and $\mu = 1.83 \times 10^{-5} \text{ Pa s}$, respectively. However, mucus has strong non-Newtonian fluid behavior. Therefore, viscosity μ is computed as a function of shear rate $\dot{\gamma}$ by employing the Power Law model:

$$\mu = \begin{cases} \mu_{\min} & \dot{\gamma} > \dot{\gamma}_{\max} \\ k \dot{\gamma}^{n-1} & \dot{\gamma}_{\min} < \dot{\gamma} < \dot{\gamma}_{\max} \\ \mu_{\max} & \dot{\gamma} < \dot{\gamma}_{\min} \end{cases} \quad (8)$$

with lower and upper shear rates, $\dot{\gamma}_{\min}$ and $\dot{\gamma}_{\max}$, respectively, and the coefficients $k = 4.6314 \times 10^{-3}$ and $n = 0.7757$. The surface tension coefficient of mucus is set to $\sigma = 50 \times 10^{-3} \text{ N/m}$.

The convective term in the momentum equations is discretized with a second-order Linear Upwind Stabilized Transport (LUST) scheme. The remaining terms (Laplacian, gradient, and interpolation) are discretized with second-order Gauss linear schemes. Temporal discretization is second-order backward. The time step size is automatically adjusted to achieve a constant Courant number of 0.2. The turbulence is modeled using Implicit Large Eddy Simulation (ILES), which resolves large flow scales, while smaller ones are modeled by the spatial discretization of the convective terms in the momentum equations.

Geometry and mesh

The geometry of the computational domain was adopted from Zhao et al. [24] (see black lines in Fig. 3, left). They employed a time-dependent shape of the glottal orifice as follows:

$$r_w(x, t) = \frac{D_0 - D_g(t)}{4} \tanh s + \frac{D_0 - D_g(t)}{4} + \frac{1}{2} [1 - \tanh s] \times [\beta_1 (x + CD_{\max}) - \beta_2 (x - CD_{\max})] \quad (9)$$

with the dimensionless parameter s defined as

$$s = \frac{b|x|}{D_{\max}} - \frac{x D_{\max}}{|x|}, \quad (10)$$

the maximum glottis opening of $D_{\max} = 4 \text{ mm}$, the duct height of $D_0 = 20 \text{ mm}$, constants $b = 1.4$ and $c = 0.4$, and the time-dependent glottal opening $D_g(t)$:

$$D_g(t) = \frac{D_{\max} + D_{\min}}{2} + \frac{D_{\max} - D_{\min}}{2} \cos(2\pi f_0 t) \quad (11)$$

with the minimum glottal diameter of $D_{\min} = 0.1 \text{ mm}$ and the glottal frequency of $f_0 = 142 \text{ Hz}$. The first two terms in equation (9) decide the basic shape of the vocal tract and the last term is the correction term which is responsible for the instantaneous shape of the vocal folds (i.e., convergent, divergent, closed or fully opened). Time-dependent variables β_1 and β_2 are defined as follows:

$$\beta_1(t) = \begin{cases} 0 & \text{for } t \leq \frac{1}{9}T, \\ 0.224 \left\{ 1 - \cos \left[\frac{9\pi}{4} \left(\frac{t}{T} - \frac{1}{9} \right) \right] \right\} & \text{for } \frac{1}{9}T < t \leq \frac{5}{9}T, \\ 0.488 & \text{for } \frac{5}{9}T < t \leq \frac{6}{9}T, \\ 0.224 \left\{ 1 + \cos \left[3\pi \left(\frac{t}{T} - \frac{6}{9} \right) \right] \right\} & \text{for } \frac{6}{9}T < t \leq T, \end{cases} \quad (12)$$

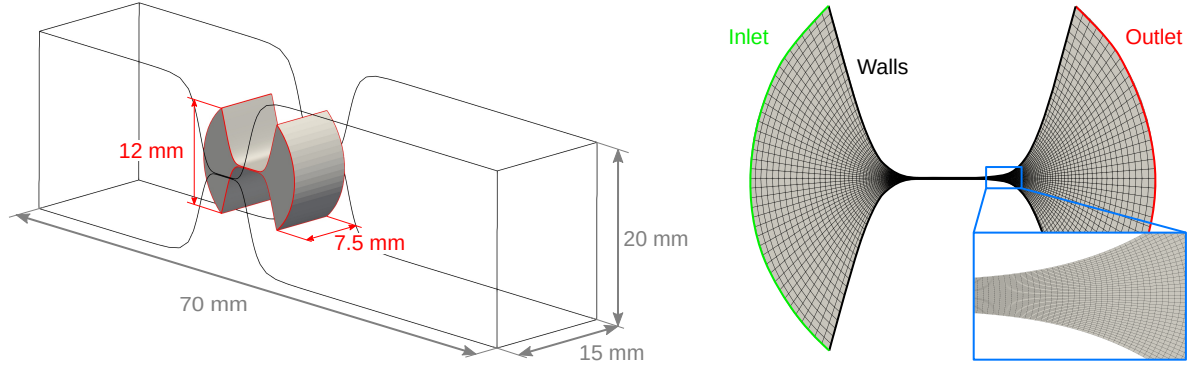


Figure 3. Left: Original geometry from Zhao et al. [24] (black) and adopted geometry of the numerical model (red); right: cross-sectional overview and close-up of the computational mesh (every other line shown).

with $\beta_2(t) = \beta_1(t + T/9)$ and $T = 1/f_0$. In the anterior-posterior direction a sine function is applied to the translational motion of the folds according to Sadeghi et al. [25]:

$$\Delta r(z) = \frac{1}{2} \left[\sin \left(2\pi \left(\frac{z}{L} - \frac{1}{4} \right) \right) + 1 \right] r(z) \quad (13)$$

with L being the length of the glottis in anterior-posterior direction.

The computational domain is reduced to the region close to the vocal folds as indicated by the red lines and gray closed surface in Fig. 3, as it is assumed that the breakup initially occurs at the center of the glottis and in order to reduce the computational cost of the simulations. Therefore, the total considered width of the vocal folds is lowered to $L = 7.5$ mm.

The computational domain consists of a structured mesh with a mesh resolution of $644 \times 80 \times 800$ cells in streamwise, wall-normal and spanwise direction. This results in a total cell count of 41.2×10^6 . The structured topology of the mesh automatically results in a high mesh resolution at the glottis and a coarser mesh at the inlet and outlet boundaries. The actual cell size varies throughout the simulation as the glottis is opening. It ranges from $12 \mu\text{m}$ to $30 \mu\text{m}$ during the formation of the liquid bridge and its breakup, respectively.

Based on Scherer et al. [26], a total pressure of 1250 Pa is defined at the inlet with a static pressure of 0 Pa at the outlet. The walls are considered no-slip and a perfectly wetting mucus layer with a uniform thickness of $100 \mu\text{m}$ is initialized at the start of the simulations. Based on the frequency f_0 , it takes 7 ms for a complete period of opening and closing of the vocal folds. Since the breakup of the liquid bridge occurs during the opening, only 2.4 ms are required to capture this effect, though.

Results and discussion

Experimental results

Overall, the high-speed recordings of the VF during oscillation reveal four different types of mucus breakup. Figure 4 shows one representative image for each of these observed droplet-generating mechanisms.

Figure 4(a) presents the mechanism that is characterized by a liquid film spanning the complete glottis. This film is inflated by the exhalation airflow, forming a large bubble with a necking area. One side of this bubble expands faster than the other and hence ruptures first. Surface tension then causes the rim to retract quickly, tearing apart the less inflated side of the bubble as well. As a result, ligaments develop along the rim of the punctured bubble and generating numerous very small particles.

The burst of a small bubble, spanning only half of the glottis while leaving the other side exposed, is shown in Figure 4(b). Similar to the bursting of the large bubble, when the small bubble ruptures, the ligaments of a receding

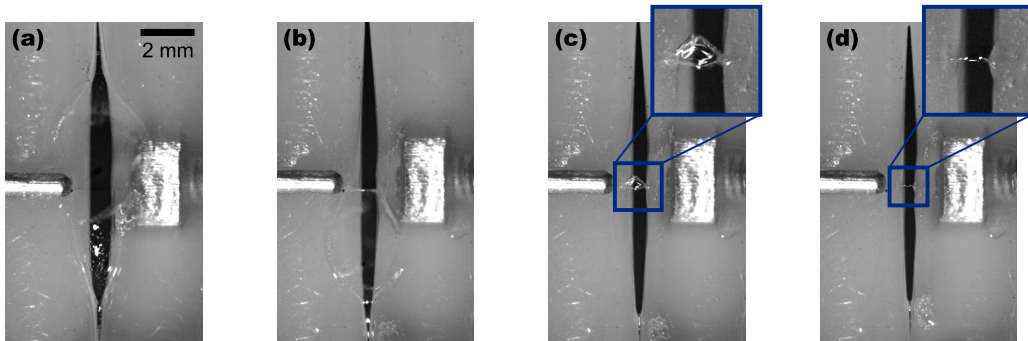


Figure 4. Different droplet generation mechanisms observed in the model experiment (top view of the vocal folds in upstream direction): (a) breakup of large bubble, (b) breakup of small bubble, (c) jetting liquid bridge breakup, (d) filament breakup.

rim break into tiny droplets. However, small bubbles also generate larger droplets, primarily at their thickened edge that connects the VF over the center of the glottis.

Figure 4(c) illustrates the phenomenon we call "jetting". As the VF start to separate, two opposing surfaces of a liquid film accelerate toward each other, resulting in breakup into droplets due to the introduction of instabilities when they collide at the center. Jetting generates larger droplets than bubble burst, as a single compact liquid volume is fragmented. Occasionally, jetting can also be observed at the edges of small bubbles.

Inhomogeneities in the mucus layer thickness can lead to a thin liquid bridge, which is constricted into a fine filament as the glottal gap widens (see Fig. 4(d)). Capillary pressure induces further thinning of the filament [27]. These filaments are unable to form a single large satellite droplet as the aerodynamic forces of the perpendicular airflow cause additional stretching and instability. Instead, the filament breaks up into a number of droplets. Filament breakup also produces larger droplets compared to bubble bursting because the filament's diameter is significantly greater than the bubble film thickness.

Often, some or all of the mechanisms described above occur in alternation over the course of a measurement (one parameter combination, several subsequent oscillation periods). However, there are also cases, where only one of the mechanisms appears. Table 1 lists all studied parameter combinations. The breakup mechanisms occurring in the respective cases are indicated as well. Two parameter combinations were only considered in the shadowgraphy measurements for particle size evaluation. Whether and which of the aerosol generation mechanisms occur will provide further information on why the droplet size distributions are shaped in a certain way.

Table 1. Summary of all investigated parameter combinations (air flow rate $\dot{V}$, frequency f , amplitude $\hat{x}$). Shadowgraphy measurements to evaluate particle size distributions were conducted for all of the cases. For the cases where also atomization process visualization was performed, the observed droplet generation mechanisms (B...large bubble; b...small bubble; J...jetting; F...filament) are listed.

$\dot{V}$ (l/min)	f (Hz)	$\hat{x}$ (mm)	Breakup Mechanisms
10	100	0.7	B
20	100	0.7	b, J, F
30	100	0.7	b, J, F
20	75	0.7	b, J
20	125	0.7	B, b, J, F
20	150	0.7	
20	100	0.5	
20	100	1.0	B

The PDFs of the particle size distributions obtained from the shadowgraphy measurements are presented in Figure 5. A sample of, on average, about 4300 droplet size values contributes to each distribution, ensuring the statistical reliability of the obtained results. For the assessment of the findings, it is important to note that the imaged droplets can be considered as being still in completely undehydrated (wet) conditions, as the spray was recorded very close to the VF. After being emitted into the ambient air, mucus droplets quickly dehydrate, shrinking to droplet nuclei with an equilibrium diameter of only about 20 to 40 % (depending on the amount of non-volatile matter) of their initial size, enabling them to stay suspended in the ambient air for a much longer period of time [28, 2]. Due to this strong reduction in size, even initially quite large particles can form persistent aerosols [29]. All particle size distributions measured in this study have in common that the first and highest peak is always located at diameters of about 14 μm . Consequently, the majority of the particles created here will shrink to sizes of about 2.8 to 5.6 μm , allowing them to remain airborne for up to 100 min [28].

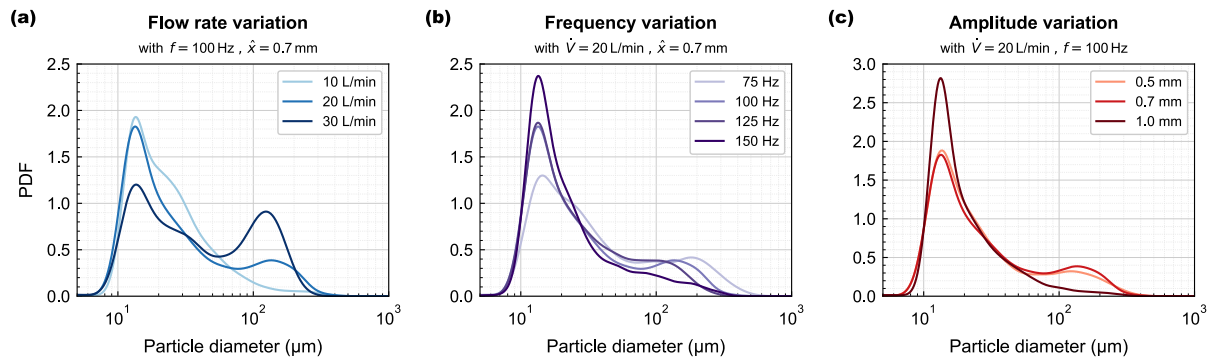


Figure 5. Probability density functions (PDFs) of the particle size distributions for (a) comparison of different flow rates at constant frequency 100 Hz and amplitude 0.7 mm, (b) different frequencies at constant flow rate 20 l/min and amplitude 0.7 mm, (c) different amplitudes at constant flow rate 20 l/min and frequency 100 Hz.

Almost all of the PDFs represent bimodal distributions, which suggests a connection with the different droplet formation mechanisms that were discussed before. No parameter variation influences the position of the first peak, however, the emergence and position of a second peak are strongly dependent on the setting of the exhalation parameters. The first peak of all measured particle size distributions develops at the small diameter of approximately $14\text{ }\mu\text{m}$, whereas the position of the second peak varies in the range of large particle sizes of $100\text{--}200\text{ }\mu\text{m}$.

According to Figure 5(a), an increasing flow rate leads to the emergence—no second peak is present at the lowest flow rate 10 l/min —and a strong growth of the second peak from 20 to 30 l/min while maintaining the position at about $130\text{ }\mu\text{m}$. Concurrently, the height of the first peak is being decreased. These results align with the findings from visualizing the atomization mechanisms (cf. Table 1). Specifically, at the lowest flow rate, only the burst of large bubbles is observed, whereas at higher flow rates, both bubble burst (small and large) and also filament breakup and jetting are occurring. Consequently, increasing flow rates generate a significantly higher amount of larger droplets and reduce the proportion of smaller particles.

Contrary to the flow rate dependencies, increasing the frequency (see Fig. 5(b)) leads to smaller particle sizes. Similar to the distributions observed for flow rate variations, second peaks appear within the diameter range of $100\text{--}200\text{ }\mu\text{m}$. However, as the frequency increases, the peak shifts toward smaller particle sizes without a change in height. While all distributions for frequencies between 75 and 125 Hz exhibit this second peak, it disappears at 150 Hz , causing a significantly larger proportion of very small particles. This suggests that as the vibration frequency increases, the mucus is subjected to higher velocities and accelerations, resulting in the breakup into smaller droplets.

Lastly, varying the oscillation amplitude as depicted in Figure 5(c) reveals yet another behavior. At both smaller amplitude values (0.5 and 0.7 mm) the PDF curves are almost identical, with further increase of the amplitude there is a discontinuity and the second peak disappears completely, as the burst of large bubbles is the only persisting droplet generating mechanism. This results in considerably more very small particles. In general, higher amplitudes correspond to speaking louder. Bagheri et al. [8] also found a higher proportion of particles with a wet size of $\mathcal{O}(10\text{ }\mu\text{m})$ during louder speech, which corresponds to our first peak in the PDFs.

From the comparison with the visualized atomization mechanisms, it is clear that the first peak is caused by the bursting of bubbles. A second peak only arises when other mechanisms such as filament breakup and/or jetting are also present. It becomes evident that when the PDFs are unimodal, presenting an especially high amount of very small particles, only the "large bubble burst" mechanism is observed during atomization process visualization (see Table 1). This implies that the presence of bubbles extending across the entire glottis hinders other mechanisms that could generate larger droplets, such as jetting and filament or small bubble breakup. The substantial increase in the proportion of small droplets of $\mathcal{O}(10\text{ }\mu\text{m})$ is particularly significant in terms of pathogen transmission, as these aerosol particles, which are large enough to carry various pathogens, can linger in the surrounding air for an extended period and potentially accumulate. The order of magnitude of the diameters of the sampled droplets is consistent with what would be expected for the laryngeal mode based on literature sources [30, 8, 31]. Nevertheless, it should be noted that the image resolution of the camera poses a limitation that makes it impossible to detect particles smaller than $5\text{ }\mu\text{m}$ and additionally causes an absolute error of $\pm 5.2\text{ }\mu\text{m}$ for the deduced diameter values.

Simulation results

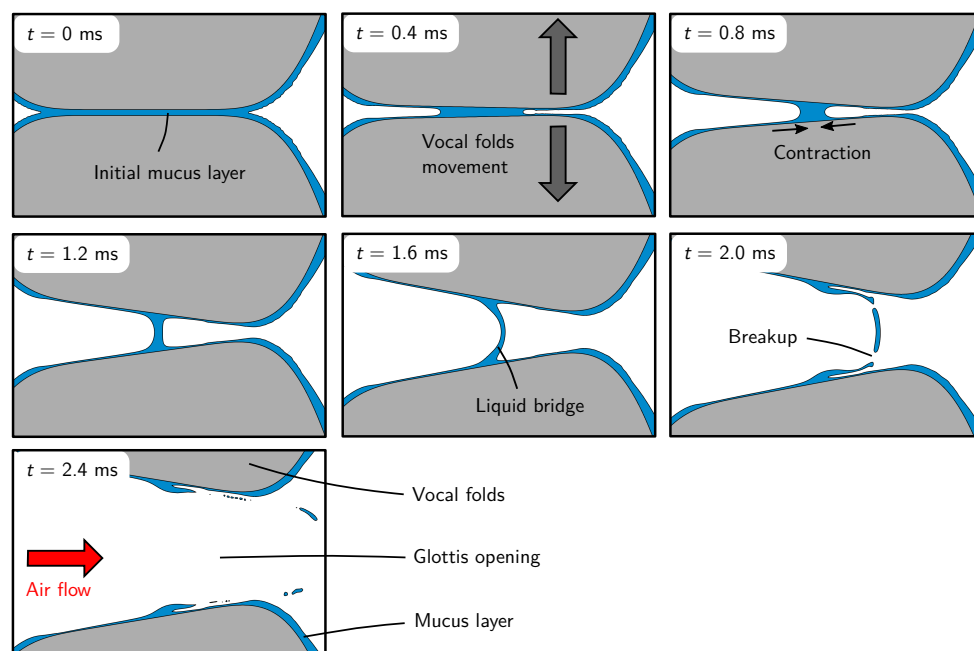


Figure 6. Mucus layer (blue) in a cross-sectional view through the computational domain of the vocal folds: Movement of the vocal folds, development and breakup of the liquid bridge.

Figure 6 shows the vocal folds in a cross-sectional view through the computational domain. The different stages of liquid bridge formation and breakup are recognizable. Initially at $t = 0$ ms, the vocal folds are considered closed with a minimum opening of $100\text{ }\mu\text{m}$, which is completely filled with mucus. As the vocal folds open and the mucus contracts due to conservation of mass, a liquid bridge forms. Its thickness decreases over time ranging from 1.3 mm at $t = 0.4\text{ ms}$ to $120\text{ }\mu\text{m}$ at $t = 1.2\text{ ms}$. After 1.6 ms , the thickness stays roughly constant at about $75\text{ }\mu\text{m}$ and a deformation of the liquid bridge is recognizable due to the aerodynamic forces resulting from the pressure difference between inlet and outlet. This deformation ultimately leads to the rupture of the liquid bridge visible at 2 ms and the remaining mucus gets transported further downstream with the airflow.

The cross-sectional view of the liquid bridge formation and breakup shows only a simplified representation of the overall process. This becomes apparent when visualizing the mucus layer in streamwise direction from downstream the vocal folds as shown in Figure 7. At 1.6 ms , several bubbles of the liquid bridge have already burst. These openings in the mucus lead to a pressure equalization between upstream and downstream region resulting in a jump in air flow velocity. As this process continues, numerous ligaments develop, which eventually break up and form smaller droplets further downstream.

Droplet generation mechanisms observed in experiment and simulation are very similar, as breakup of (small) bubbles and filament breakup are observed in both. This is despite the perfect idealized conditions present in the numerical model: (1) The simulation captures only a single vocal folds opening period, (2) it starts with a homogeneously initialized, perfectly wetting mucus layer, and (3) there are no external disturbances to the flow except from the vocal folds movement, such as upstream turbulence.

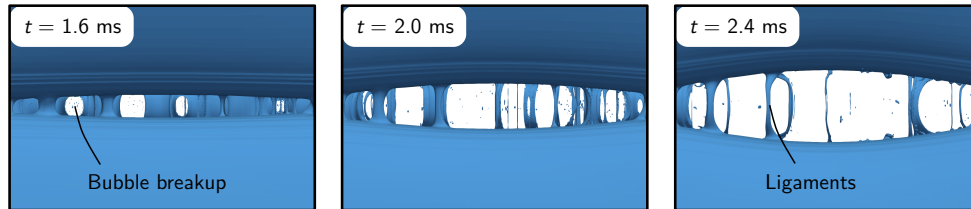


Figure 7. Formation and breakup of liquid bridges viewed from the outlet in upstream direction.

Due to the non-Newtonian rheology of mucus, its viscosity varies greatly with shear rate. In order to estimate the present viscosity, it is vital to obtain information about the resulting shear rates in the mucus layer during opening of the vocal folds. Therefore, Figure 8 depicts the shear rate for both mucus and air in a cross-sectional view through the computational domain. Highest shear rates with up to $1 \times 10^6\text{ 1/s}$ occur within the air phase upstream and downstream the liquid bridge. In the mucus, however, the shear rates are much smaller. Within the liquid bridge, values of $(10^3 - 10^4)\text{ 1/s}$ are reached while in the mucus layer itself, maximum values are in the range of 10^2 1/s .

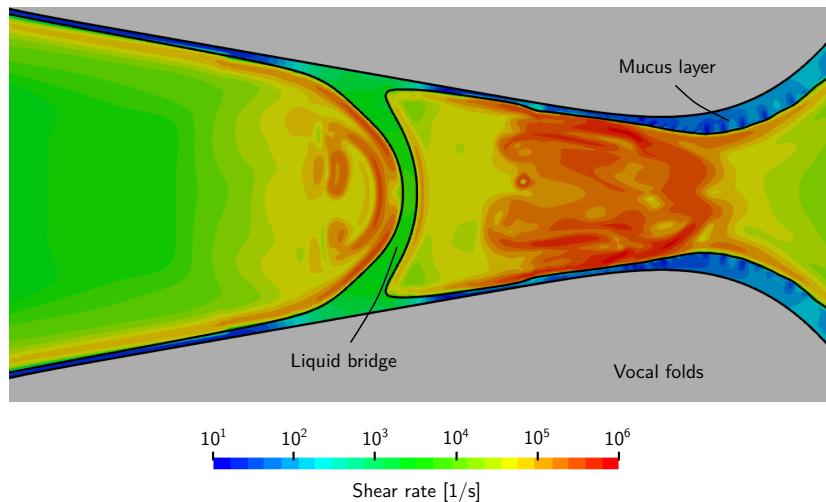


Figure 8. Shear rate in a cross-sectional view through the computational domain of the vocal folds at $t = 1.6\text{ ms}$.

Conclusions

This pioneering study experimentally and numerically investigates aerosol formation at simplified models of the vocal folds. It provides a unique perspective since aerosol generation at the vocal folds—particularly the distribution of particle sizes—can be viewed in isolation from other sources such as the lungs and mouth/nose, which would be impossible with in vivo measurements.

The atomization processes of respiratory mucus were visualized through both in vitro model experiment and 3D multiphase CFD simulation, which revealed reproducible yet diverse mechanisms. The breakup of bubbles formed

in the glottal gap was found to dominate the aerosol generation process. Experimental observations and simulation results exhibit considerable similarities regarding the extension, inflation and breakup of liquid films within the glottis. This study was already able to identify tendencies regarding the parameter dependencies of particle size distributions originating from the vocal folds. Increasing the air flow rate leads to the generation of a higher proportion of large particles, while increasing either frequency (pitch) or amplitude (loudness) results in a higher amount of small particles. The atomization at the experimental model vocal folds produces particles in the size range expected during human voicing. Through dehydration, these particles are able to form long-lasting aerosols.

For future studies, we plan to obtain artificial mucus with more realistic properties, improve the vocal folds models, and expand them to include the upper respiratory tract. Additionally, to identify parameter settings that result in a high number of exhaled particles, it is necessary to measure the particle concentration or emission rates alongside the particle size distribution.

Overall, these findings provide valuable insights into aerosol formation during phonation, and further research can build upon these results to enhance our understanding of aerosol generation and dynamics in the respiratory system.

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Nomenclature

3D	three dimensional	α	phase fraction (–)	L	glottis length (m)
CAD	computer aided design	β	time-dependent variables (–)	$\mathbf{n}$	interface normal vector (–)
CFD	computational fluid dynamics	$\dot{\gamma}$	shear rate (1/s)	p	static pressure (Pa)
COVID	coronavirus disease	μ	dynamic viscosity (Pa s)	r_w	shape of glottal orifice (m)
CSF	continuum surface force	ρ	fluid density (kg/m ³)	s	dimensionless parameter (–)
KDE	kernel density estimator	b, c	constants (–)	T	period duration (s)
MFC	mass flow controller	D_0	duct height (m)	t	time (s)
LED	light emitting diode	D	glottis opening (m)	$\mathbf{u}$	fluid velocity (m/s)
LUST	linear upwind stabilized transp.	f	frequency (Hz)	$\dot{V}$	flow rate (m ³ /s)
ILES	implicit large eddy simulation	$\mathbf{f}_\sigma$	surface tension force (N)	$\hat{x}$	amplitude (m)
PDF	probability density function	$\mathbf{g}$	gravitat. acceleration (m/s ²)	x, z	spatial coordinates (m)
RH	relative humidity	k	free surface curvature (1/m)		
VF	vocal folds				

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